

WEEK - 8

Identifying Various Types and Components of an Academic Report

Video 1: Introducing Academic Report Writing: Definition and Purpose

Definition of an Academic Report

An academic report is a specific type of academic writing to communicate

- Analyses of literature
- Laboratory experiments
- Investigations on issues
- Project findings

The information is presented in a formal, concise, and objective manner

Academic reports are written for various readers or audience

- Academics
- Legal practitioners
- Corporate managements
- Government organizations
- Customers or clients

To write an effective report, it is important to keep the readers/audience in mind.

Purpose of an Academic Report

- Present the analysis of research findings, scientific experiments, case studies, surveys, etc.
- Provide the readers/audience with the rationale for the research or experiment, or survey that has been done
- Present the analysis objectively
- Create the scope for further studies and help other researchers to work further on a specific topic

Structure of an Academic Report

Title

- The specific topic of the report

Acknowledgments

- To briefly thank those who helped in providing and compiling information/data for the report

Table of contents

- A list of main sections with corresponding page numbers

Abstract

- The main aspects of the report that help readers know if it is relevant to their research

Introduction

- The background, context, purpose, and scope of the report

Literature review

- An analysis of the literature relevant to the research

Methods

- The research design and how the research has been done

Results/discussion

- A summary of the results or findings of the research

Conclusion

- The conclusion of the research that has been drawn based on the analysis or experiments

References

- A list of sources that have been used and mentioned in the report

Video 2: Different Types of Academic Reports

Letter Report

- A letter report is a type of brief report written in the form of a letter.
- Usually used for property, legal, and business-related issues.

Letter Report : Components

Heading

- Must have the return address, the contact information of the sender, and the date.

Address of the recipient

- Should be correct and complete, including the recipient's formal position.

Salutation

- Should be formal
Examples: "Dear (name)" or "Mr./Ms./Mrs. (Title)"

Body

- Has three parts-introduction, findings, and recommendation.

Complimentary close

- Includes a polite remark that ends the letter
Examples: "Best Regards," "Thank You," "With Appreciation," etc.

Signature line

- Describes the sender's full name

Laboratory Report

Video subtitle:

“Then coming to the Laboratory Report. So, suppose you were working on certain kind of experiment. And you're getting a lot of data and data set. And then you are reaching to certain kind of result discussions. So, how ideally you should present that?”

Laboratory reports are written

- To explain scientific concepts
- To analyze laboratory experiments

Laboratory Report: Components

Title

- Describes the experiment that has been conducted.

Abstract

- Mentions the summary of the experiment and introduces the purpose and results.

Introduction

- Includes the background of the study, the motivation behind the experiment, the relevant theory, and the research hypothesis.

Methods and Materials

- Discusses the steps taken to conduct the experiment, including the equipment and apparatus.

Results

- Reflects the findings of the experiment and is represented through tables and figures.

Discussion

- Analyzes the findings, strengths, and weaknesses of the experiment design.

Conclusion

- Summarizes the whole process, findings, and significance of the study and offers scope for further research.

Research Report

Video subtitle:

“Research reports are written to present an in-depth analysis of a systematic research. The purpose, process, and results are research are clearly discussed in this type of reports. And this also has a kind of a structure which is a bit of similar to lab report but not exactly the same the way lab reports are done.”

Research reports are written to

- Present an in-depth analysis of systematic research.
- Clearly describe the purpose, process, and results of the research.

Types of research reports

- Quantitative research report
- Qualitative research report

Research Report : Components

Summary

- A brief overview of the research, its purpose, and objective.

Introduction

- Introduces the research, the research questions, and the research hypothesis, if any.

Methods

- Mentions the steps taken to complete the research.

Results

- Includes the research findings.

Discussion

- Analyzes the research results.

Conclusion

- Presents a summation and new understanding of the research topic.

References

- Includes the list of sources mentioned in the research.

Video 3: Writing Styles and Formats of Academic Reports

Styles of Academic Reports

The format of writing the report depends on which style of report you are writing.

Formats of three styles of academic reports

1. Research or lab report
2. Conference report
3. Inspection report

Research or Lab Report

- Used to record an experiment or a study
- The focus is on facts, analysis of results, and describing the methodology

Format	Introduction
	Methodology
	Results
	Analysis and discussion
	Conclusion
	References and appendices

Conference Reports

- Written by an attendee or by a member of the organizing committee.
- Convey information such as theme, sub-themes, host institution, speakers, date, and location.

Format	Statement of purpose
	Proceedings and key information
	Learning achieved and reflections on the experience
	Acknowledgments to funding bodies or persons of importance

Video subtitle:

“ Now coming to the conference report. Conference reports are written by an attendee or by a member of the organizing committee. It conveys the information such as theme, sub-themes, host institution, speakers, date and location, and it also has a structure specific somewhat which usually follows the following format, so it started a statement of purpose, so you give an idea, so what's the relevance of this confidence and where it intervenes in our existing knowledge base like scholarship, another thing. Then the proceedings and the key information which is view discuss about the exact events and related and relevant information and data. Then learnings achieved and reflections on the experience, so how these are the things which we gained from a conference and how they could have been used in near future, so this is the thing, so the learning goal where they are at the beginning and then how we achieve the goal to in this conference and the reflections on the experience, so this is a part. Finally, the acknowledgements to funding bodies and persons of important, so the resource persons who funds the confidence, who held the confidence to proceed in a certain way, you should be formally, officially, professionally with a proper acknowledgment to that setup person.”

Inspection or Performance Reports

- To measure if an expected standard or benchmark has been achieved.
- For example:
 - National Assessment and Accreditation Council (NAAC) report on various universities
 - Appraisal reports by managers

Format	Purpose of assessment
	Frameworks or rubrics used
	Summary of findings with evidence
	Results
	Recommendations

Video subtitle:

“ It usually follows the following format, so the purpose of assessment why this assessment was done, so what is the purpose, what are the purposes for instance and frameworks are rubrics you, so what are the criteria on which certain institutions, universities and organizations where like evaluate it, like could be infrastructural. It could be academic achievement, since we're talking about an act reports could be in fact structural. It could be academic achievement. It could be a larger purpose like what their studies, their experiments, their events like what purpose they had and finally the summary of findings with evidence, so here it's very important that you are writing a formal report which will be used for a professional poppers, so you also provide certain evidence that in datasets sudden an which elaborate the point erasing, and it also would make the report more serious and more professional and then result and finally the recommendation, so depending on the achievements or the set of a result, you would have certain recommendation at the very end of inspection or the performance report.”

Video 4: Different Components of Academic Reports**Components of Academic Reports**

These components are common among most types of reports.

- Title page
 - Gives key information about the academic reports
 - Title
 - Writer's name
 - Designation
 - Institutional affiliation
 - Date of submission
- Abstract
 - Provides a short summary of the academic report that gives an overview of all parts of the report
 - Helps the reader decide whether to read the entire report
- Table of contents
 - Includes a list of all the sections and subsections in the report
 - Provides page numbers for each section/subsection
- Introduction
 - Talks about the purpose of the particular report
 - Presents the research objectives
- Literature review
 - Talks about the available research, such as previous studies related to the report in question
- Methods
 - Mentions how the data collection was conducted
 - Provides a justification for the same
- Results and discussion
 - Presents the results of the experiment or experience, or assessment
 - Discusses its implications
- Conclusion

- Provides a summary of all the key findings discussed in the report
- Provides recommendations or suggestions for the future
- Reference section
 - Lists out all the other works, such as books, journal articles, and websites, which were referred to by the author while writing the report

Video subtitle:

“Very important note is that each of every section will be very important when you write an academic report that you should keep in mind. Every section has its own design and own structure that also you need to keep in mind.”

Writing Introduction Section of an Academic Report

Video 1: Structure of the Introduction Section

Function of Introduction in Academic Reports

- Introduction
 - Has a specific function in academic reports
 - Introduces the study and places the study in the research literature of a particular field
 - Provides information about the perspectives available in the field and specifies the perspective adopted in the study
 - States the theoretical framework of the study
 - Mentions the current research trends in the field of study
 - Defines the scope of the study and clearly states what is not included in the study

Structure of an Introduction



Introduction: Examples

1. Artificial intelligence (AI) systems powered by deep neural networks (DNNs) are pervasive across society. (General statement)
2. The surveys cover a large body of work that may prove hard to navigate and synthesize into a broad view of the field. Instead, this a article investigates a simple space of foundational explainable DNN methods. (Objective of the study)
3. They run in our pockets on our cell phones (Georgiev et al., 2017), in cars to help avoid car accidents (Jain et al., 2015), in banks to manage our investments (Chong et al., 2017), and evaluate loans (Pham & Shen, 2017). (Relevant literature)

Video subtitle:

“Now, let us look at an example of how an ideal an introduction section of an academic report could look like. First come into the general statement. It can start like artificial intelligence or AI system powered by deep neural networks or DNNs are pervasive across society. It is somewhat general light so this could be your first sentence. You can start with an introduction with these sentences. Then coming to the objective of the study. Then you can write the surveys cover a large body of work that may prove to navigate and synthesized into a broad view of the field. Instead, this article investigates a simple space of foundational, explainable DNN methods. Clearly you're saying this is what you did and this is what your idea of doing that not the entire broad field. Now, relevant literature. It could like state something like that. They run in our pockets on our cell phones, in cars to help avoid car accidents, in banks to manage our investments and evaluate loans. When we talk about or write about specific points, you can see the way it's structured. We're also providing the existing scholarship, so that's surname and the publication year and then you are focusing on the relevant literature. This is one idea, one structure you could take up when you write an introduction section.”

Video 2: Different Components of the Introduction Section**Essential Components of an Introduction****1. Introducing the research objectives**

Example: In this paper, we, for the first time, integrate motifs into MASs and propose a Motif-Aware Weighted Multiagent System (MWMS) framework for consensus control (Xie et al., 2021).

2. Briefly stating the relevant literature and the research gap

Example: At present, motifs are mostly used for analyzing network structure in biology (Kim & Choi, 2009), engineering (Milo, et al. 2002), social media (Coletto, et al. 2017) and statistical relational learning (Das, et al. 2019). However, there is still a lack of work studying and leveraging motifs in MASs. According to our experiments, the introduction of motifs to consensus protocols in MASs can achieve better results (Xie et al., 2021).

3. Presenting the findings of the study

Example: In our experiments, we compare the traditional and proposed algorithms under the same initial topology network. The results show that the convergence effect of the proposed MWMS framework for MASs is better, i.e., the splitting of the system can be effectively reduced by introducing motif into the control function (Xie et al., 2021).

4. Presenting the structure of the paper

Example: The rest of this paper is organized as follows. In section 2, some definitions of symbols and necessary knowledge are given to facilitate the following model description: Section 3 contains the construction of the MWMS framework and its application in multi-agent consensus (Xie et al., 2021).

Optional Components of an Introduction

1. Opening the introduction with a general statement

Example: Distributed Artificial Intelligence (DAI) mainly studies how logically or physically dispersed intelligent systems solve problems in parallel and in cooperation with each other (Xie et al., 2021).

2. Presenting state-of-the-art research or theories in the field of study

Example: DAI can be divided into two basic research areas: Distributed Problem Solving (DPS) and Multi-Agent Systems (MAS). As an important field of DAI, MASs have attracted an increasing amount of attention from experts in computer science, control theory (Dobbe, Fridovich-Keil, and Tomlin, 2017), cooperative, engineering, reinforcement learning (Lin, Adams, and Beling, 2019; Wai, Yang, Wang, and Hong, 2018), etc.

Video subtitle:

“So these are the major important components of your introduction section academic report. When you write an academic report, if you keep in mind this is the structure there's an important component, then obviously your introduction will be more clear and the receiver will get a proper clarity of your findings in the research. That's why this is very important to remember all these key points when you structure introduction section of your academic report.”

Video 3: Literature Review: Definition and Purpose

What Is Literature Review?

- Reading and reviewing research works in the respective discipline
- Also includes:
 - Understanding the study objective
 - Theoretical framework employed
 - Data type
 - The methodology followed
 - Data collection tools used
 - Findings of the study
 - Inferences made from the findings
- Includes objective evaluation of the study in terms of theoretical framework, methodology, findings, and interpretation of the findings
- Compares the objectives, theoretical frameworks, methodologies, findings, and interpretation of the findings between studies
- Studies in the fields are classified and grouped according to themes that can be reported in different paragraphs

Techniques Used in Literature Review

- Quotation:
 - directly quoting the extracts from the original source. It must be used judiciously in a research report.

- Example: Kate's (2002) definition of mob violence 'organized or spontaneous act of physical assault against individuals or specific groups motivated by ideology or hatred' (p.9).

Video subtitle:

"First is quotation, like when you're using an excerpt from certain given source in a research report, if certain words and phrases and ideas and key arguments which are very important, very much relevant to your research, then maybe you can quote them. But then directly quoting the extract from the original source is like may not be a very good idea for every possible sources. And when you do the quotation and like a kind of you should be very careful about the citation of the particular researcher and the scholarship that you mentioned here."

Video subtitle:

"But as I said, quoting certain work directly may not be a very good idea. It shows a kind of a less effort from your part so that you are directly quoting rather than summarizing the thing. And sometimes in your literature review you read a lot of materials which are related. And then if you start quoting every sources, the report looks clumsy and untidy and not a very serious kind of research report. So you should be very careful so that you don't randomly put quotation rather what is the preferred mode is that we summarize the literature which is existing and then you write the findings and the arguments and the opinion of the other people, but in your own words."

- Summarizing
 - To systematically synthesize the commonalities in different research studies.
 - Example: Many efforts have been made in developing decentralized consensus algorithms, such as hybrid multi-agent systems (Zheng, Ma, and Wang, 2018; Nebel, Bolander, Engesser, and Mattm uller, 2019), co-ordination problem of multi-agents (Bhargava, 2019; Li, Yang, Cai, Yang, and Wang, 2019), the consensus of multi-agent systems (Xu and He, 2018; Motsch and Tadmor, 2014)..
- Paraphrasing:
 - To paraphrase the points from relevant work.
 - Example: Steels (1998) explored mechanisms by which a group of agents forms a vocabulary and ontology in a self-organized way through local interaction and without central control authority. His focus was on robotic agents that share the same sensorimotor capabilities and which, by means of language games, develop their own shared ontology, creating and grounding their vocabulary on the perception of a shared physical space.
- Comment:
 - To present the authors' view on a work or a theory or group of works.
 - Example: Our techniques do require that agents can participate in many interactions, particularly when vocabularies are large. Nowadays, however, obtaining these rehearsal interactions is likely cheaper than crafting a linguistic or ontological resource that is useful to obtain alignments.

Purpose of Literature Review

- Place the research study in the literature
- Avoid repeating the work and make the work original
- Express our understanding of the field
- Establish the gap in the research literature
- Contribute to the field by addressing the gap in the research literature

Video 4: Writing the Literature Review Section

Collecting, Summarizing, and Synthesizing Information from Sources

- Identify the argument in each source or work
- Collect specific information from each source regarding the theoretical framework, methodology, and findings
- Compare the information gathered from each source
- Track the different streams of arguments and frameworks from the sources
- Use synthesis grid to classify the themes from sources:

	Source 1	Source 2	Source 3	Source 4	Source 5
Theme A					
Theme B					

Information-Prominent vs. Author-Prominent Referencing

Two common types of referencing in the literature review section

- Information-prominent referencing
 - Used to highlight the information
 - Author's information will be backgrounded
 - Example: The impact of smoking is high among Indians because of the moderate living conditions (Muthu, 2015; Prakash, 2016; Sen, 2019).
- **Video subtitle:**
 - “So you are referring the author's research and contribution in that field, but you're not starting with their name or their ,kind of, prominence in the particular sentence. Rather, your idea is more to engage with the information in a kind of a more serious way and then you were referring the scholarship. Author-prominent referencing on the other hand is used to highlight the contribution of the author.”
- Author-prominent referencing
 - Used to highlight the author's contribution
 - It indicates the importance of the author's contribution to the current study
 - Example: Littlejohn (2006) has studied the impact of smoking on the quality of sleep.
- **Video subtitle:**
 - “If the researchers uses author-prominent referencing, it indicates the importance of the author's contribution for the current study.”

Organizing the Literature Review Section

Organizing the literature review depends on the researcher's intent

	When the researcher...
Chronological	Wants to highlight the field's evolution
Thematic	Has identified recurring themes in the field
Methodological	Wants to highlight how using different methodologies has provided varied results
Theoretical	Categorizes research works based on the theoretical framework

Video subtitle:

“ So four ways of organizing literature reviews are generally were to use in our regular writing of introduction section are chronological, thematic, methodological, and theoretical. Chronological is used when the researcher wants to highlight the evolution of this field. Thematic is used when the researcher has identified recurring themes in the field. Methodological is used when the researcher wants to highlight how using different methodology has provided varied results. Finally, theoretical is used when the researcher wants to categorize research works based on the theoretical framework. This is a kind of a way you can organize them, and then sometimes there could be a kind of overlap between them. Chronological and thematic, or thematic and methodological, depending on the requirement of your field of studies and area of research, and what kind of academic report you're writing. So, these are some ideas that you should keep in mind, some of the techniques and strategies you should keep in mind when you write your introduction section and particularly the literature review of that section.”

Writing Methods Section of an Academic Report

Video 1: Structure of the Methods Section

Methods Section and Its Significance

- The methods section of a report refers to:
 - The equipment and apparatus
 - The literature used to conduct an experiment
 - The research on which the report is written
 - It offers significant information on the validity of the research
 - It is required to clearly state the procedure of how an experiment or research was done and the rationale behind determining that procedure
- The methods section includes:
 - The steps that were taken to answer the research questions
 - Offer a description of how the research was done
 - Justify the research design
 - Provide a description of how the findings were studied

Structure of the Methods Section

- Explanation
 - Each subsection of the methods section states
 - The underlying principles behind the research
 - Rationale behind choosing the tools to conduct the research
 - Offers a detailed explanation of all these components
- Meticulous description of details
 - Each step of the research has to be mentioned meticulously in the methods section.
 - The research design has to be clearly outlined, providing justification for the design choice.
 - The reliability and validity of the selected research tools depend on the research design.
 - The research design has to be described in detail.
- Ethical approval
 - The method section should clearly mention that the measures adopted for the study have ethical approval from the concerned authority.
- Specifications of the variables
 - The control, dependent, and independent variables must be analyzed and specifically mentioned in the methods section
- Visual representations
 - The methods sections can include visual information to add clarity to the description of the research design

Methods vs. Methodology

Research Methods	Research Methodology
1. Refers to the tools and steps adopted to conduct a research	1. Includes an analysis of those steps and provides an explanation for choosing them
2. Refers to how the research/experiment is conducted	2. Explains the rationale behind the research/experiment
3. Comprises different techniques to conduct the research	3. Includes the entire strategy that can be adopted to complete the research
4. Used to find solutions to research problems	4. Outlines the correct process that can be followed to determine the solutions
5. The scope is narrower than the research methodology scope	5. Has a wide scope and dimension

Video 2: Different Components of the Methods Section

Components of Methods Section

- Research design
 - Refers to the strategy adopted to answer the research questions
 - States the justification for adopting a specific strategy for a specific research
 - Often included in the first paragraph of the methods section
- Setting
 - For research that includes surveys, fieldwork, or even laboratory experiments
 - The methods section must justify the rationale behind choosing a specific setting
- Ethical approval
 - Includes the ethical approval for a research
 - An institutional review committee or a regulatory body must approve the research
 - It is a significant component in the methods section
- Participants
 - Depending on the type of research, participants/samples/subjects are determined
 - The methods section offers the rationale behind this choice
- Equipment
 - Equipment and apparatus are essential for laboratory-based research
 - Surveys or interviews are often mandatory for research on humanities and social sciences
 - These are instruments for the study that the methods section describes

- Procedure
 - A detailed description and analysis of the procedures taken while conducting the research
 - Clearly outlines
 - The nature of the participants
 - The criteria for selecting samples
 - The justification for the literature choice
- Data collection
 - An effective research tool for both qualitative and quantitative research
 - The methods section mentions:
 - Details about the data collected
 - The justification behind determining the type of data collected
 - The analysis of how the data provides answers to research questions
- Data analysis
 - Followed by data analysis
 - Its subsection often includes statistical analysis as well

Important Points to Remember

- Reviewing samples of relevant methods section from relevant journals provides an effective understanding of the sample collection process or participant's selection.
 - **Video subtitle:**
 - “So reviewing samples of relevant method section from journals related to the field of study provides an effective understanding of the process of sample collection or participant's selection. So you should be careful while reading certain things and also taking a decision out of that reading. So this method and why not that method.”
- There are specific procedures for collecting data. Being acquainted with the procedures creates convenience and makes the research organized.
 - **Video subtitle:**
 - “There are specific procedure for collecting data. Being acquainted with the procedure creates convenience and makes the research organized. So you need to spend less time when in the field if you have a proper plan which is at the design and decide, okay, this is what will be my research plan then.”
- An understanding of the literature relevant to the field of research is required for conducting effective research
- The subsections and information in the subsections of the methods section should be presented in a logical and comprehensive manner.
- The researcher should avoid redundancy as redundancy can result in incomprehensibility.
 - **Video subtitle:**
 - “The researcher also should avoid redundancy as redundancy can result in incomprehensibility. So just decide on what is your method section will have and what is not important, not necessary here. So you can simply just remove those parts and only include the important part in the method section. So this will make your

methods section more precise and also to the point, and more clear, and it would also indicate some serious kind of an effort from you.”

Video 3: Describing Tools and Materials

Describing Tools or Materials

- Forms a part of the methods section
- Clearly describes the various tools or materials used by the researcher and the purpose of its use
- Provides clarification on the choice of particular tools or materials
- May compare the tools or materials used to

Describing Tools or Materials: Example

Py, Cherif, and Habet (2022) describe their tools as follows:

Current Max-SAT solvers are able to efficiently compute the optimal value of an input instance, but they do not provide any certificate of its validity. In this paper, we present a tool called MS-Builder, which generates certificates for the Max-SAT problem in the particular form of a sequence of a equivalence-preserving transformations. Additionally, we propose an extendable tool called MS-Checker, which is able to verify the validity of any Max-SAT certificate using Max-SAT inference rules. Both tools are evaluated on the unweighted and weighted benchmark instances of the 2020 Max-SAT Evaluation.

Video subtitle:

“So you are clearly giving us an idea how we were using new tools and the relevance, and the importance. And how you can also say why this is a certain kind of tool you're using, and not the other state of. “

Describing Tools or Materials: Do's and Don'ts

- Do's
 - Describe each tool separately and concisely
 - Include physical tools like questionnaires and equipment used
 - Mention versions and sources of software used
 - Use appropriate sub-headings or bullet points to improve readability
 - Use appropriate citations to refer to tools or materials created by other researchers
- Don'ts
 - Don't Include standard software and lab equipment such as Microsoft word or a keyboard
 - Don't Give long descriptions of tools or materials if already available in referenced sources

Video 4: Describing the Process of Data Collection

Data Collection

- Allows us to answer research questions and prove hypotheses
- Includes description of various kinds of data gathered from different sources
- Helps give validity to conclusions and generalizations
- It can be done through a variety of methods:
 - Surveys
 - Questionnaires
 - Observations
 - Measurements
 - Interviews
 - Focused group discussions
 - Monitoring databases or forums
 - Tracking online sources

Describing Data Collection

Video subtitle:

“Describing data collection process. There are certain kinds of questions which should be very clear at the beginning of your data collection process and you should be very careful that you are providing answer to this set of questions.”

- What was the purpose of I collecting the data?
- Who performed the collection of data?
- Where did the data collection take place?
- When or over what time period did the data collection take place?
- What were the different kinds of data collected?
- How does the collected data help answer the research question?
- What modes were used to collect the data?
- What tools were used to collect the data?
- How was the data stored?

Data Collection: An Example

The data used in this study was collected from Fair Health Services, which works with 14 hospitals across the state. All the patients who consented to participate were seen over a period of two weeks at the respective hospitals. Medical professionals collected data such as demographics, vitals, medications taken over the previous year, and results of laboratory tests, which were updated on a secure, shared database. This data was then used for statistical modeling of patient outcomes based on current patient descriptors to improve the medical decision-making process.

Video subtitle:

“You are giving a clear clarity of your data collection method. You are giving an idea of sample you have chosen and how you collected them, and over what period you collected them and also how you're storing them, what kind of database and other set of information which is very important for this part.”